

Operating Systems (CSC-217)
Credit Hrs. 2 (Pre-Req : Data Structures)
Google Classroom Code Section A: zk4hz3e5
Google Classroom Code Section B: gxyzqofu
Google Classroom Code Section C: kv4e7kc5
Contact Hours: Mon & Wed (9:30 am – 11:00 am)

Semester: **Fall 2025**
Instructor: **Noreen Khalid**
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Course Objective:

This course covers history and goals of an operating system. Among the topics covered will be the cutting-edge advances in computing that are redefining operating systems, the structure and responsibilities of operating systems, design considerations, such as performance, fault tolerance, security, modularity and cost. Recent operating system design issues arising from the rapid growth in distributed computing, made possible by the Internet and the World Wide Web will also be discussed.

Program Learning Outcomes (PLOs):

- PLO1: Academic Education
- PLO2: Knowledge for Solving Computing Problems
- PLO3: Problem Analysis
- PLO4: Design/Development of Solutions
- PLO5: Modern Tool Usage
- PLO6: Individual and Team Work
- PLO7: Communication
- PLO8: Computing Professionalism and Society
- PLO9: Ethics
- PLO10: Life-long Learning

CLO–PLO Mapping with BT Levels

CLO	Learning Outcome	BT Level	Mapped PLOs
CL01	Understand the characteristics of different structures of the Operating System and identify the core functions of the Operating System.	C2	PLO1, PLO2
CL02	Analyze and evaluate the algorithms of the core functions of the operating system and explain the major performance issues with regard to these functions.	C4, C5	PLO2, PLO3, PLO4
CL03	Demonstrate the knowledge in applying system software and tools available in modern operating systems.	C3	PLO4, PLO5, PLO7

Recommended Books: Soft Copies available

A. Operating System Concepts by Abraham Silberschatz, Peter B. Galvin, and Greg Gagne, Wiley; 10th edition

B. Operating Systems: Internals and Design Principles by William Stallings, Prentice Hall; 9th Edition

C. Operating Systems: Three Easy Pieces

Course Outline and Schedule:

Week No.	Topic	Lecture No.	Lecture Contents	Relation with CLO	Assessment's
W1. 22-Sep	Introduction & Background	L1.	– Introduction to operating system – Computer System Overview, Application software, System software, Microprogramming, Physical devices – Evolution of Operating Systems – Types of Operating Systems – Computing Environments – History of Operating Systems, Introduction to Unix, MS-DOS and Windows	CL01	
W2. 29-Sep	Operating System Structure	L2.	– Process management, Memory management, File management, I/O system management, Secondary storage management, Networking – System protection – Operating system services – Operating System Structure, OS layered approach, OS/2 layer structure, Microkernel System Structure	CL01, CL02	
W3. 06-Oct	Operating System Structure	L3.	– Operating System Design and Implementation – System design goals, Mechanisms and policies, Operating system implementation – Traditional UNIX System Structure, Microkernel System Structure, Solaris Modular Approach, Mac OS X Structure, Android Architecture – Operating System Generation (SYSGEN) – Kernel/Kernel Mode – System Calls, System Call Processing, Types of System Calls – Interrupts, Interrupt Processing, Types of Interrupts	CL02, CL03	Assignment 1
W4. 13-Oct	Process Concepts	L4.	– The process concept, Program vs process – Process creation, Process control block, Process table, Shell, Operation on processes – Kernel/kernel mode – System calls, Types of systems calls – Interrupts, Interrupt processing, Types of Interrupts	CL02, CL03	Class Test 1

W5. 20-Oct	Process Scheduling	L5.	– Context Switch, Process states, Process states transitions, Process life cycle, five state process model, Unix process state transition diagram – Scheduler, Scheduling algorithm, Objectives of Scheduling, Criteria for scheduling, CPU-bound vs I/O-bound processes – Types of Scheduling, Process scheduling queues – Scheduling Algorithms, FCFS, SJF, Priority, Round Robin, Multilevel feedback queues scheduling – BSD Unix scheduling, Multiple processor scheduling – Threads, Thread support Solaris	CL02,3	Assignment 2
W6. 27-Oct	Memory Management	L6.	– Memory, memory hierarchy, Process loading and swapping – Memory management, Memory allocation methods, Single partition allocation – Storage management strategies, Fetch strategies, Placement strategies, Replacement strategies, Variable partition with compaction	CL02,3	Class Test 2
W7. 03-Nov	Memory Management	L7.	– Non-contiguous memory allocation – Paging vs Segmentation – Simple paging, Implementation of paging, Page address translation, Implementation of Page Table, Paging with TLB – Page Table Structure, Hierarchical Page Tables, Hashed Page Tables, Inverted Page Tables – Simple segmentation, Segment addressing, Paging vs. Segmentation – The Intel IA-32 Architecture, Logical to Physical Address Translation in IA-32, Intel IA-32 Segmentation, Intel IA-32 Paging Architecture – ARM Architecture	CL03	

W8. 10-Nov	Virtual Memory	L8.	– Virtual memory, Mechanism of virtual memory – Translation Look Aside Buffer, Address translation, Page fault – Page replacement, Page replacement algorithm, FIFO, Belady's anomaly, Optimal LRU, Random page replacement, LFU, MFU, Non-used recently page replacement algorithm – Paging Mechanism, Additional techniques, Memory protection and sharing – MS-DOS Memory management, Windows memory management, Unix memory management	CLO3	
W9. 17-21 Nov	Mid-Term	L9	– Exam		Mid-Term 1
W10. 24-Nov	Threads & Multitasking	L10	– Multitasking, Threads – Types of threads, Kernel threads, User level threads – Multithreading model, One to One Model, Many to One Model, Many to Many Model – Solaris 2 threads, Windows 2000 threads, Linux threads, Java threads	CLO3	Assignment 3
W11. 01-Dec	Concurrent Processes	L11	– Concurrent processes, Resources, Race condition – Synchronization, Critical section, Mutual Exclusion, Deadlock	CLO3	Class Test 3
W12. 08-Dec	Inter-process communication	L12	– Pipes – Semaphores, use of semaphore for critical section problem – Monitors – Signals, Message passing, use of messages for synchronization – Shared memory, Object linking & embedding – Client-Server Communication, Sockets, Remote Procedure Calls (RPC), Remote Method Invocation (RMI)	CLO3	

W13. 15-Dec	Input / Output Management	L13	– Input/Output organization, Objectives of I/O system – Buffering, Double buffering transfer, Multiple of circular buffering transfer – Unix I/O system, MS-DOS I/O system, Device drivers in windows – Disk structure, Seek time, Latency – Disk scheduling, FCFS, SSTF, Scan scheduling	CLO3	Assignment 4
W14. 22-Dec	File Management	L14	– File Types, File Identification – MS-DOS, UNIX, Windows file naming – Alias Filenames – File management techniques (MS-DOS, Unix, Windows) – Unix Inode pointer system, Unix volume structure	CLO3	Class Test 4
W15. 29-Dec	Distributed operating systems	L15	– Distributed System, Distributed Operating System, Distributed OS vs Network OS – Distributed System's Design Issues, Distributed System Transparencies, DOS Information Management Solution – Distributed File System, DFS Structure – File Naming Schemes, Remote File Access – Statefull vs. Stateless Service, Statefull File Service, Stateless File Server, File Replication	CLO3	
W16. 05-Jan	Project Presentations	L16	– Project Presentation		
W17. 10-13 Jan	Quiet Week		Quiet Week		Quiet Week
W18. 14-24 Jan	Final exam		– Final exam		
29 Jan	Exam Scripts Shown		–		
30 Jan	Declaration of Result		–		

Assessment Criteria (100%)

Assessment Description	Weight (%)	CLOs	PLOs
Quizzes (at least 3)	10%	CLO1, CLO2, CLO3	PLO1, PLO2
Assignments (at least 3)	10%	CLO1, CLO2, CLO3	PLO1, PLO2
Project / Presentation	10%	CLO3	PLO4, PLO5, PLO7
Mid Term Exams	25%	CLO1, CLO2, CLO3	PLO2, PLO3, PLO4
Final Exam	45%	CLO1, CLO2, CLO3	PLO2, PLO3, PLO4

GRADING TABLE ABSOLUTE SCALE if no. of students in Final Exam are up to 20, otherwise RELATIVE SCALE			
Range	Percentage	Points	Grade
85.00-100	85.00	4.00	A
81.50-84.99	81.50	3.67	A–
78.00-81.49	78.00	3.33	B+
74.50-77.99	74.50	3.00	B
71.00-74.49	71.00	2.67	B–
67.50-70.99	67.50	2.33	C+
64.00-67.49	64.00	2.00	C
60.50-63.99	60.50	1.67	C–
57.00-60.49	57.00	1.33	D+
50.00-56.99	50.00	1.00	D
0.00-49.99	Below 50	0.00	F